

MAT2007 - Introduction to programming - Project

Smoking Prevalence and Cigarette Consumption

Who is Winning the Fight against Nicotine?

A global analysis looking at smoking decline by sex from 1980 to 2012

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Introduction

With a shocking number of more than 8 million deaths related to tobacco each year, smoking remains one of the biggest health threats to modern society, leading the WHO to call it the global tobacco epidemic (Organization, 2023). The most commonly used form of tobacco is cigarettes; however, all forms of tobacco are harmful and are a leading cause of cardiovascular and respiratory diseases, as well as a multitude of cancers.

Although smokers are aware of these risks and health concerns, the number of smokers has increased with the growth of the population. In contrast, the smoking cessation rate remains low, with an estimate of only 9.6% of adults 18 years of age and older who successfully quit.

Reportedly, a higher percentage of females quit (10.8% vs. 8.9%), motivating the question of whether prevalence also declined faster among females than males ("Cancer Trends Progress Report", 2025). Therefore, the following data analysis describes the differences in prevalence decline by sex, following the question: **Has smoking prevalence declined faster among females than among males between 1980 and 2012, in 187 countries?**

Materials and Methods

Data

The dataset "Global Smoking Prevalence and Cigarette Consumption 1980-2012" from the Institute of Health Metrics and Evaluation (IHME) contains data covering smoking prevalence (the proportion of the population smoking at a certain time, in percentage), the number of daily smokers, cigarette consumption, and annual rate-of-change estimates separated by country and by sex, recorded over 30 years (1980-2012), providing data for 187 different countries ("Global Smoking Prevalence and Cigarette Consumption 1980-2012 — GHDx", 2021). Estimated data were sourced from numerous prevalent studies, surveys, articles, national reports, and ministries of health websites, as well as databases like the WHO Global Infobase. For further information on the surveys included in the data set per year and country, see supplementary Table 15 of Ng et al. (2014).

The data subset used for this analysis includes the prevalence of smoking in females and males. In addition, the dataset provides uncertainties for each sex as upper and lower bounds, which describe the 95% confidence interval of the estimates.

Data Analysis

This data analysis was performed using Python in Visual Studio Code, including the following packages: *os* (to create an output folder in the working directory), *Pandas*, *NumPy*, *Matplotlib* (to plot the graphs in Figure 1), *SciPy* (to perform the paired t-test) and *Seaborn* (for the graph outline).

Subsets of the dataset were created that contain the year, country, and prevalence of females and males.

For each year, the mean prevalence was calculated in all countries for females and males, plus the standard deviation (SD), the number of countries (n) and the standard error ($SE = SD/\sqrt{n}$) (Appendix A Table 3). Additionally, the confidence intervals of the means were calculated and used as shaded bands in Fig. 1a and 1b.

After, the relative decline was calculated by taking the 1980 mean of each sex as a baseline, converting the yearly means to % change from that baseline, to create better comparability between the female and male prevalence curves. For each country, a linear regression was fitted to the separate percentage changes over time for females and males using *linregress* from *SciPy*. The regression slope represents the average annual change as a percentage of the country's 1980 prevalence.

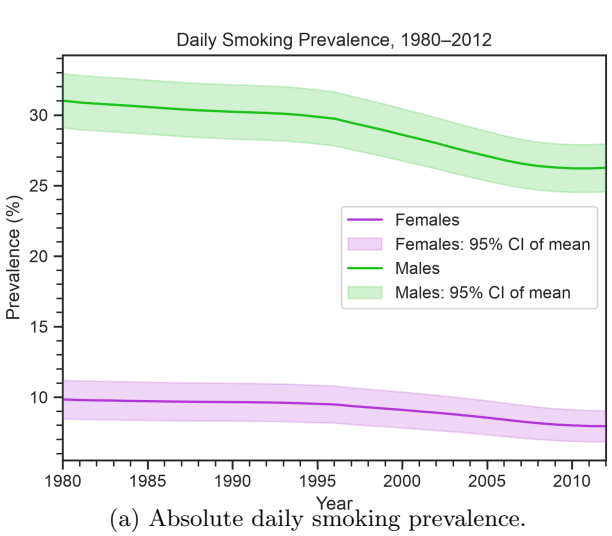
Lastly, two one-sample t-tests and one paired t-test were performed using *ttest_1samp* and *ttest_rel* from *SciPy*. They tested whether prevalence declined significantly for each sex, and whether the female and male slopes differed within the same country (significance level $\alpha = 0.05$).

Results and Discussion

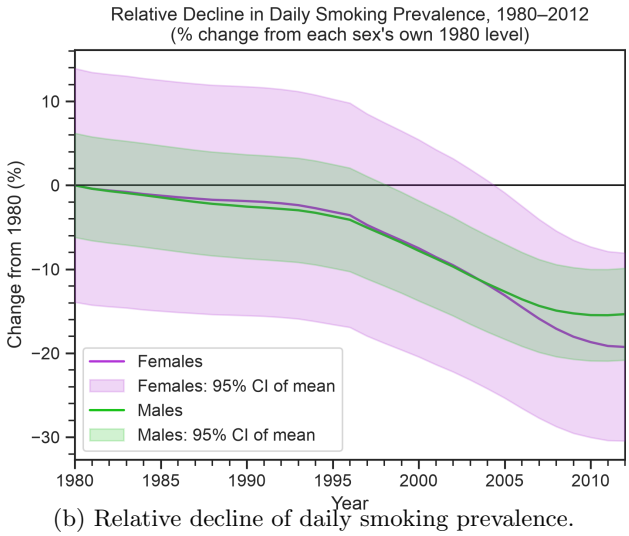
Figure 1a shows the mean prevalence of smoking every year in all 187 countries, calculated separately for females and males, with shaded bands describing the 95% confidence interval (CI) of the mean.

Male prevalence fell from 31.0% to 26.3% (−4.8 percentage points), and female prevalence from 9.8% to 7.9% (−1.9 percentage points). Because males start from a much higher level, their absolute drop is larger by construction, so absolute prevalence cannot show which sex declined faster. This graph alone is not enough to state who is winning the fight against nicotine.

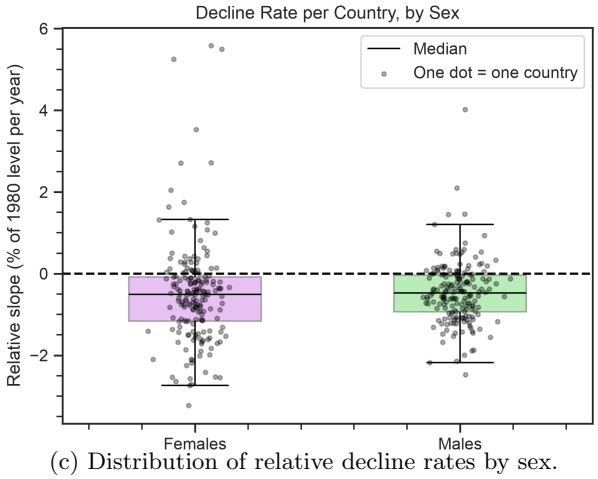
Figure 1: **Daily smoking prevalence by sex across 187 countries, 1980–2012.**



(a) Absolute daily smoking prevalence.



(b) Relative decline of daily smoking prevalence.



(c) Distribution of relative decline rates by sex.

Smoking Prevalence by sex across 187 countries (1980-2012). (a) Unweighted mean daily smoking prevalence (%) across countries for females (purple) and males (green). (b) Means expressed as percentage change from each sex's own 1980 mean. In (a) and (b), shaded bands show the 95% CI of the mean across countries (mean ± 1.96 × SE). (c) Each dot is one country's slope of relative change in prevalence (% of 1980 level per year); boxes show the median and interquartile range, with whiskers extending to the most extreme countries within 1.5×IQR. Data: Ng et al. (2014).

To address the baseline issue Figure 1b displays the same yearly means but converted to percentage change from each sex's own 1980 mean, with a horizontal line at zero marking no change as a point of reference and the shaded 95% CI. In relative terms, the mean prevalence across countries declined by approximately 15% for males (31.0% to 26.3%) and by approximately 19% for females (9.8% to 7.9%), as shown by the gap between the two curves in Figure 1b.

Both curves fall below zero, showing a relative decline in smoking prevalence for both sexes. What is clear is the proportional progress for both sexes towards a more negative percentage change from their respective

1980 baselines. The curves almost overlap until about 2005, after which the female curve is slightly lower, and both flatten from about 2010, seemingly plateauing. The 95% CIs overlap almost completely, so the two curves cannot be distinguished. The female band is wider because the female SD (9.554) is almost as large as its mean (9.837) in 1980. Although there is a visually observable gap between the two curves, the prevalence of both sexes decreased by a similar relative amount between 1980 and 2012; the difference of about 4 percentage points lies within the overlapping confidence bands"

To test whether prevalence declined at all and whether it declined at different speeds in the two sexes, each country was reduced to its relative slope at the country level. The slope is the average change per year, expressed in % of the 1980 level per year. This gives 187 slopes for females and 187 for males. Figure 1c shows these slopes as a distribution for each sex, where each dot represents one country, spread sideways so the dots do not overlap (horizontal position has no meaning). The box covers the middle 50% of countries, the black line inside the box is the median, and the whiskers extend to the most extreme countries within 1.5×IQR of the box. Any point below the reference line at zero marks a country where the prevalence fell, and any point above is a country where the prevalence rose.

Both boxes lie below or on zero, so in the typical country prevalence declined for both sexes with near-identical medians (−0.511%/year for females and −0.470%/year for males). As seen in the boxplot, the upper quartile of the male slopes is approximately zero, meaning about a quarter of countries showed no decline or an increase. This is not observed in females. Otherwise, the two sexes differ in spread, not in centre: female slopes vary more between countries than male slopes (IQR 1.084 vs 0.896; Appendix A Table 1).

Countries beyond the whiskers are mostly cases of rising prevalence. The most extreme increases are found in females in Tonga, Greece, Lithuania, Portugal, and Bulgaria. The male outliers are in Kazakhstan and Saudi Arabia. The strongest and unusually fast decline in the entire dataset occurred in males in Mexico, the one outlier beneath the male boxplot.

A one-sample t-test (*ttest_1samp*) was used to test the slope of each sex against zero. The null hypothesis was that, on average, prevalence did not change between 1980 and 2012. The test compares the mean slope to zero with its standard error. Country-level relative slopes were significantly below zero for both females ($t(186) = -5.30$, $p < 0.001$; mean slope −0.491%/year) and males ($t(186) = -8.46$, $p < 0.001$; mean slope −0.465%/year). The typical country therefore showed a real decline in both sexes.

Furthermore, sexes were directly compared using a paired t-test (*ttest_rel*). The test is paired because the female and male slopes come from the same country, so it analyses the difference (female slope − male slope) within each of the 187 countries. The paired comparison did not show significant differences between the sexes ($t(186) = -0.31$, $p = 0.76$; mean difference −0.026%/year, 95% CI −0.193 to 0.141). Therefore, there is no evidence that the prevalence decreased faster among females.

Figure 1c supports the conclusion of the t-tests: prevalence decreased in the typical country for both sexes, and there is no sign that one sex declined faster. All three tests are summarised in Appendix A Table 2. An important limitation to mention is that the tests describe the typical country, with all countries weighted equally regardless of population.

Conclusion

The analysis found no statistical significance in the rate of decline between females and males between 1980 and 2012 in 187 countries. Therefore, females are not 'winning' the fight against nicotine relative to males. However, it was found that daily smoking prevalence declines significantly in both females and males and that relative to their own levels in 1980, the sexes declined at similar rates. Higher cessation in females did not translate into a faster decline in smoking prevalence.

Multiple limitations of this project need to be considered. First, this analysis uses unweighted means across countries. Future work should use a population-based weighted analysis. Second, the dataset ends in 2012, before e-cigarettes and vaping became widespread. An extension of the research, covering later years and total nicotine use, is advised for a more current analysis of nicotine use and smoking prevalence. Third, IHME values are modelled estimates, not raw observations, so country trends are smoothed and not fully independent from each other.

References

Cancer Trends Progress Report. (2025, April). National Cancer Institute, NIH, DHHS, Bethesda, MD. Retrieved from https://progressreport.cancer.gov/prevention/tobacco

Global Smoking Prevalence and Cigarette Consumption 1980-2012 — GHDx. (2021). Healthdata.org. Retrieved from https://ghdx.healthdata.org/record/ihme-data/global-smoking-prevalence-and-cigarette-consumption-1980-2012

Ng, M., Freeman, M. K., Fleming, T. D., Robinson, M., Dwyer-Lindgren, L., Thomson, B., ... Gakidou, E. (2014). Smoking prevalence and cigarette consumption in 187 countries, 1980-2012. *JAMA*, *311*, 183. doi:10.1001/jama.2013.284692

Organization, W. H. (2023, July). *Who report on the global tobacco epidemic, 2023*. World Health Organization. Retrieved from https://iris.who.int/bitstream/handle/10665/372043/9789240077164-eng.pdf?sequence=1

Appendices

A Supplementary Tables

Table 1: **Distribution of relative decline slopes (% of 1980 level per year) by sex, across 187 countries.**

Statistic	Female	Male
Q1 (25th percentile)	−1.163	−0.934
Median	−0.511	−0.470
Q3 (75th percentile)	−0.078	−0.038
Interquartile range (IQR)	1.084	0.896
Lower whisker	−2.789	−2.279
Upper whisker	1.548	1.306
Whisker length (upper − lower)	4.337	3.585

Whisker bounds calculated as Q1 − 1.5×IQR and Q3 + 1.5×IQR.

Table 2: **t-tests on country-level relative slopes (% of 1980 level per year) in 187 countries.**

Test	t	df	p
Female slope vs. 0	−5.295	186	< 0.001
Male slope vs. 0	−8.457	186	< 0.001
Paired: female vs. male	−0.306	186	0.76

One-sample tests compare each sex's mean slope against 0 (no change). The paired test compares female and male slopes within each country.

Table 3: **Yearly mean, standard deviation (SD) and standard error (SE) of daily smoking prevalence (%) across 187 countries, by sex, 1980–2012.**

	Female			Male		
Year	Mean	SD	SE	Mean	SD	SE
1980	9.837	9.554	0.699	31.012	13.415	0.981
1981	9.797	9.513	0.696	30.885	13.386	0.979
1982	9.776	9.494	0.694	30.798	13.376	0.978
1983	9.759	9.477	0.693	30.726	13.377	0.978
1984	9.735	9.459	0.692	30.648	13.373	0.978
1985	9.716	9.438	0.690	30.564	13.367	0.977
1986	9.698	9.415	0.688	30.480	13.371	0.978
1987	9.682	9.390	0.687	30.398	13.377	0.978
1988	9.667	9.377	0.686	30.327	13.379	0.978
1989	9.660	9.369	0.685	30.278	13.397	0.980
1990	9.652	9.361	0.685	30.224	13.409	0.981
1991	9.643	9.345	0.683	30.186	13.425	0.982
1992	9.627	9.326	0.682	30.140	13.420	0.981
1993	9.605	9.296	0.680	30.090	13.427	0.982

	Female			Male		
Year	**Mean**	**SD**	**SE**	**Mean**	**SD**	**SE**
1994	9.569	9.263	0.677	29.996	13.405	0.980
1995	9.527	9.216	0.674	29.869	13.373	0.978
1996	9.487	9.170	0.671	29.738	13.346	0.976
1997	9.373	9.083	0.664	29.448	13.223	0.967
1998	9.280	9.017	0.659	29.174	13.116	0.959
1999	9.191	8.952	0.655	28.896	13.014	0.952
2000	9.100	8.878	0.649	28.594	12.913	0.944
2001	8.996	8.789	0.643	28.304	12.803	0.936
2002	8.901	8.716	0.637	28.009	12.681	0.927
2003	8.790	8.610	0.630	27.679	12.550	0.918
2004	8.673	8.508	0.622	27.374	12.423	0.908
2005	8.548	8.395	0.614	27.086	12.273	0.897
2006	8.410	8.271	0.605	26.802	12.111	0.886
2007	8.274	8.126	0.594	26.558	11.969	0.875
2008	8.157	7.999	0.585	26.383	11.871	0.868
2009	8.063	7.894	0.577	26.280	11.817	0.864
2010	8.000	7.793	0.570	26.217	11.794	0.862
2011	7.955	7.728	0.565	26.212	11.809	0.864
2012	7.943	7.677	0.561	26.251	11.838	0.866

Means and SDs are calculated in the 187 countries for each year; SE = SD/$\sqrt{187}$. The 95% CI bands in Fig. 1 are mean $\pm$ 1.96 $\times$ SE.